

# SapBERT Classifier Comparison

## Sections 3.4 and 3.5 | Full 768-D versus PCA-64

Encoder: SapBERT. Frozen, normalized 768-dimensional clinical-concept embeddings. PCA reduces the same embeddings to 64 dimensions. Each classifier is evaluated on both text-only and text-plus-patient inputs, producing all 12 comparisons.

Dataset: 10,000 supplied cardiac records. Fixed grouped split: 8,001 development and 1,999 test rows; labels 1-3. Every comparison uses the same rows. PCA and patient-feature preprocessing are fitted on development rows only. Classifier settings are fixed across representations.

### Text-only comparison

| Text size | Classifier             | Accuracy | Precision* | F1*    |
|-----------|------------------------|----------|------------|--------|
| 768-D     | Logistic Regression    | 60.08%   | 64.40%     | 60.63% |
| 768-D     | Hist Gradient Boosting | 60.73%   | 67.31%     | 60.33% |
| 768-D     | Random Forest          | 60.93%   | 65.67%     | 60.46% |
| PCA-64    | Logistic Regression    | 59.88%   | 64.10%     | 60.83% |
| PCA-64    | Hist Gradient Boosting | 59.93%   | 66.70%     | 59.65% |
| PCA-64    | Random Forest          | 59.53%   | 63.00%     | 59.31% |

### Text plus patient features

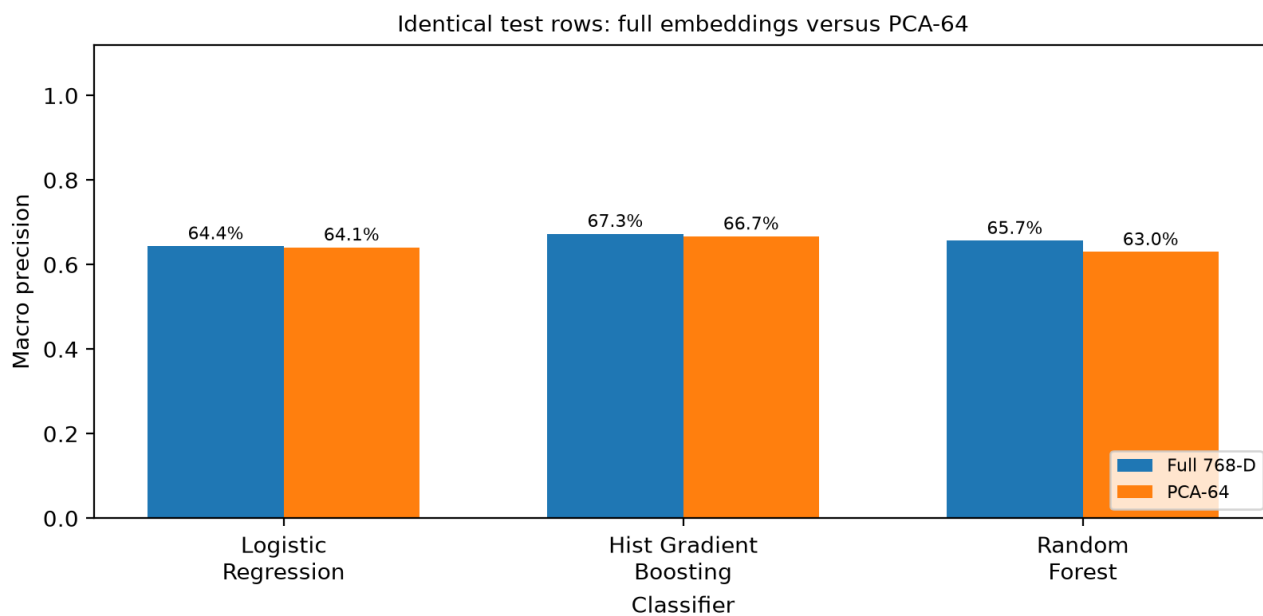
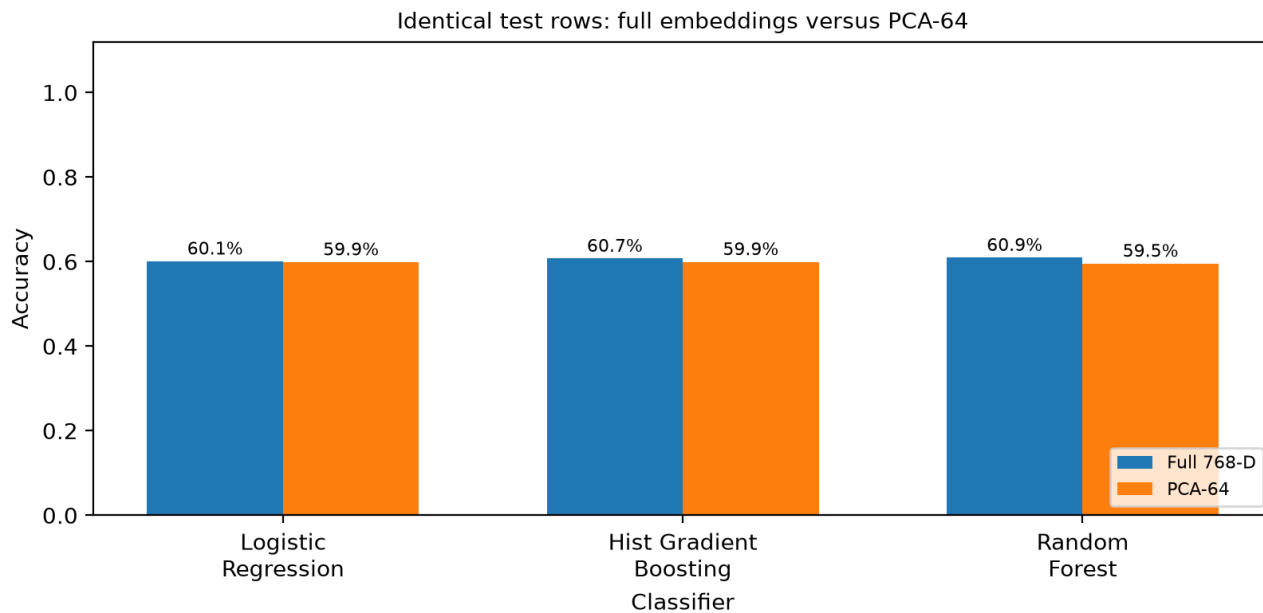
| Text size | Classifier             | Accuracy | Precision* | F1*    |
|-----------|------------------------|----------|------------|--------|
| 768-D     | Logistic Regression    | 98.50%   | 98.03%     | 98.34% |
| 768-D     | Hist Gradient Boosting | 99.10%   | 99.15%     | 98.94% |
| 768-D     | Random Forest          | 99.35%   | 99.04%     | 99.26% |
| PCA-64    | Logistic Regression    | 98.50%   | 98.03%     | 98.34% |
| PCA-64    | Hist Gradient Boosting | 99.15%   | 99.24%     | 99.01% |
| PCA-64    | Random Forest          | 99.35%   | 99.04%     | 99.26% |

\*Precision and F1 are macro-averaged. Fusion adds 22 patient features:  $768 + 22 = 790$ , or  $64 + 22 = 86$  total inputs. PCA retains 91.31% of development embedding variance.

### Interpretation

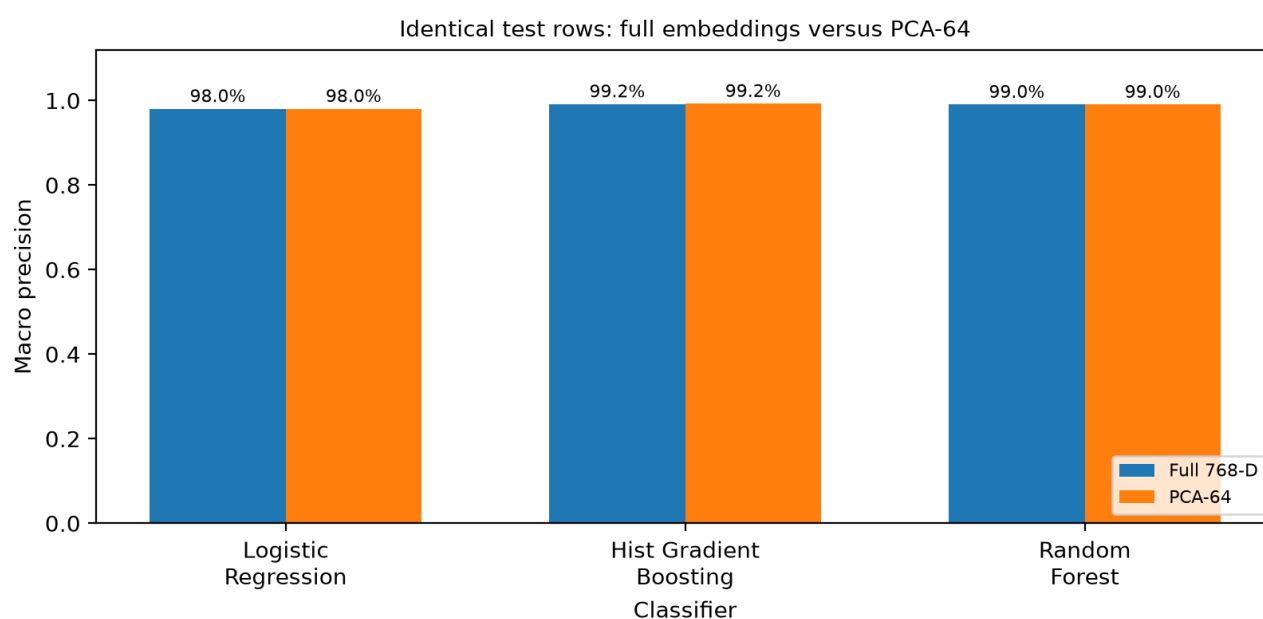
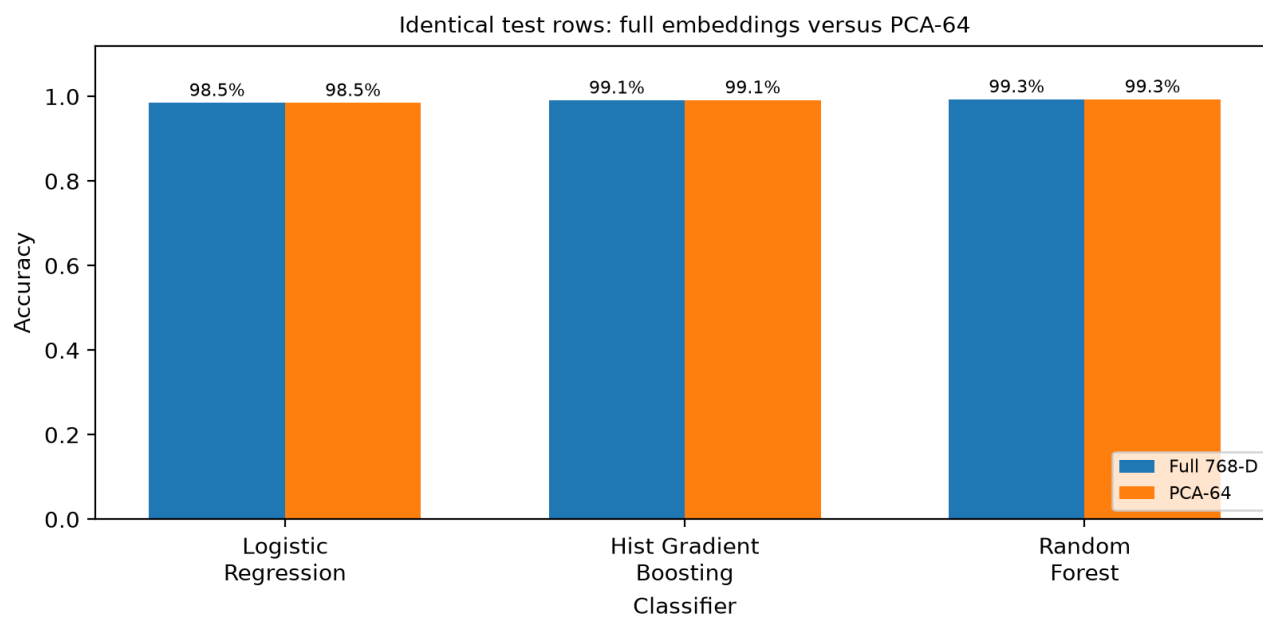
This is a fixed follow-up comparison on the existing test split, not new independent validation or test-based model selection. The dataset provider confirmed AI-assigned labels. The application model and safety checks are unchanged.

# Text-only performance



Blue bars: original 768-dimensional embeddings. Orange bars: the same embeddings reduced to 64 dimensions. Scores are from the fixed test split, not mixed cross-validation results.

# Performance with patient features



Blue bars: original 768-dimensional embeddings. Orange bars: the same embeddings reduced to 64 dimensions. Scores are from the fixed test split, not mixed cross-validation results.

# Logistic Regression: confusion matrices

Rows show reference triage levels; columns show predictions. Every matrix contains the same 1,999 test records. Top row: text only. Bottom row: text plus patient features. Left: full 768-D. Right: PCA-64.

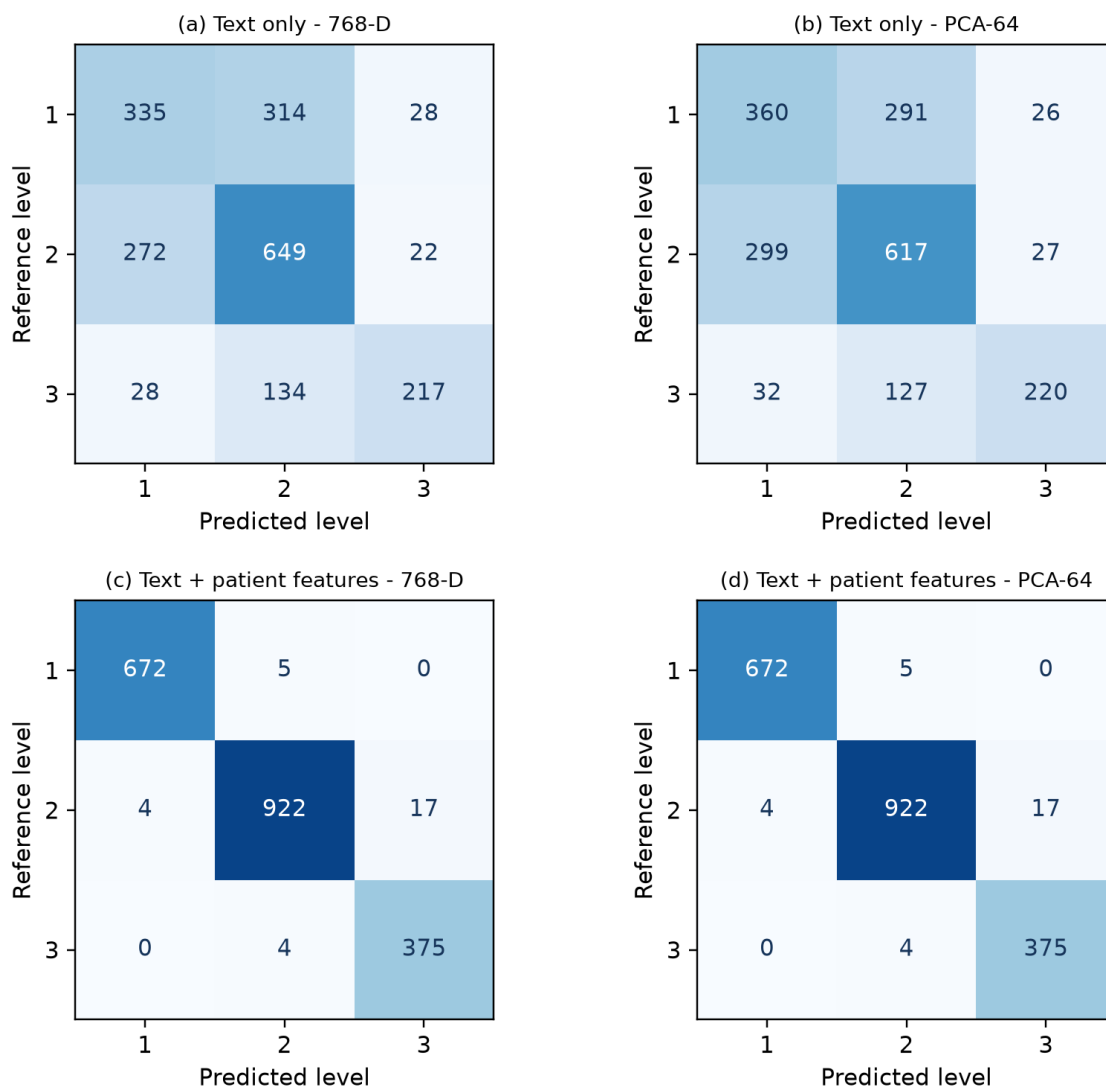


Figure: Logistic Regression under the four input conditions. Only the representation/input condition changes; the classifier settings and split stay fixed. Matrices have three levels because the supplied dataset contains no level-4 records.

# Hist Gradient Boosting: confusion matrices

Rows show reference triage levels; columns show predictions. Every matrix contains the same 1,999 test records. Top row: text only. Bottom row: text plus patient features. Left: full 768-D. Right: PCA-64.

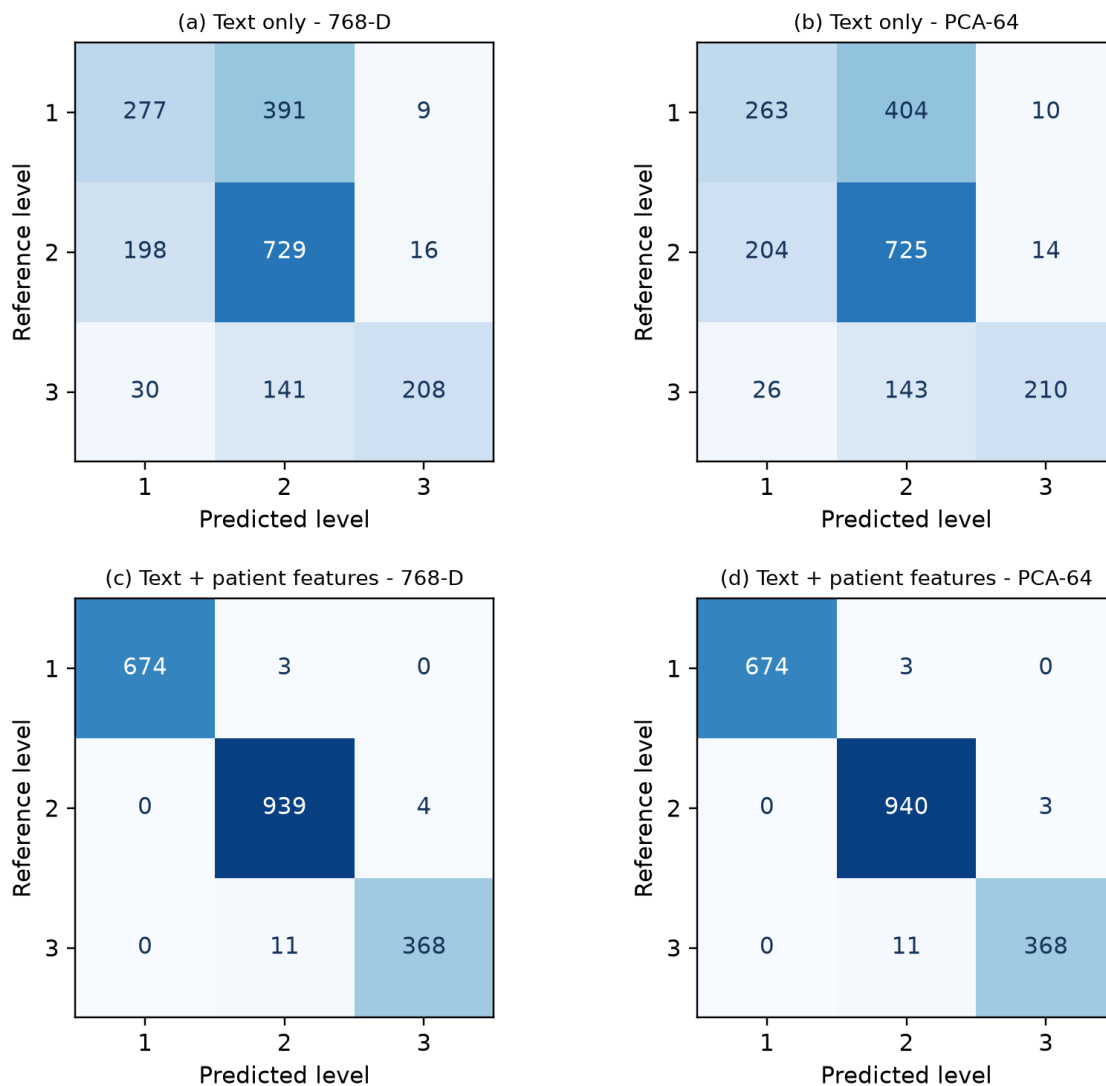


Figure: Hist Gradient Boosting under the four input conditions. Only the representation/input condition changes; the classifier settings and split stay fixed. Matrices have three levels because the supplied dataset contains no level-4 records.

# Random Forest: confusion matrices

Rows show reference triage levels; columns show predictions. Every matrix contains the same 1,999 test records. Top row: text only. Bottom row: text plus patient features. Left: full 768-D. Right: PCA-64.

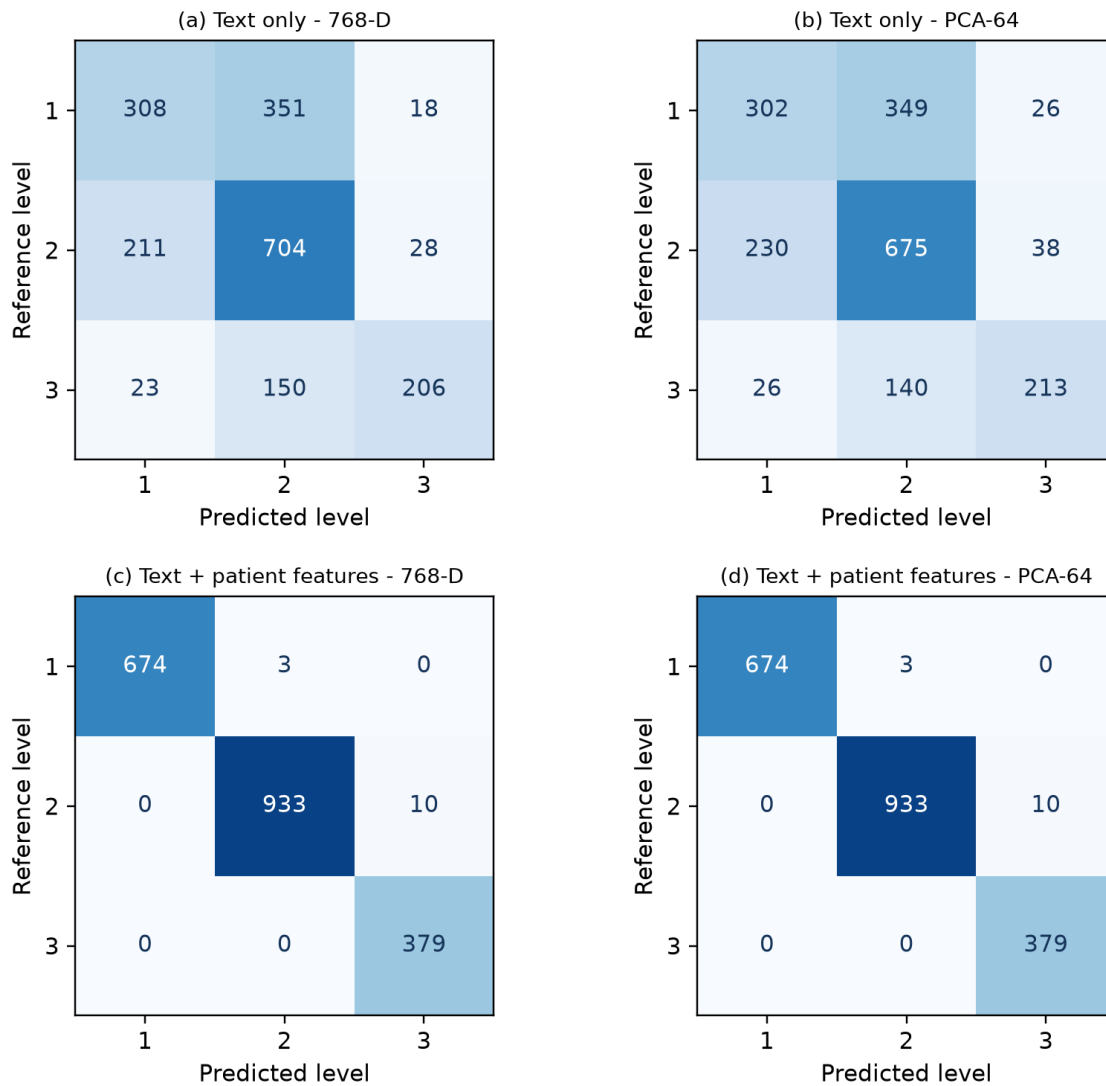


Figure: Random Forest under the four input conditions. Only the representation/input condition changes; the classifier settings and split stay fixed. Matrices have three levels because the supplied dataset contains no level-4 records.

## Section 3.6 | Comparison and reproducibility

### Performance comparison of ED triage models

| Model                                   | Accuracy | Recall | F1     |
|-----------------------------------------|----------|--------|--------|
| TF-IDF + Logistic Regression [1]        | 0.750    | 0.988  | 0.544  |
| TF-IDF + Random Forest [1]              | 0.751    | 0.964  | 0.582  |
| BiLSTM [1]                              | 0.746    | 0.846  | 0.670  |
| CNN [1]                                 | 0.723    | 0.787  | 0.667  |
| Current Logistic Regression + PCA-64    | 0.9850   | 0.9866 | 0.9834 |
| Current Hist Gradient Boosting + PCA-64 | 0.9915   | 0.9878 | 0.9901 |
| Current Random Forest + PCA-64          | 0.9935   | 0.9950 | 0.9926 |

Table layout follows the supplied screenshot. Current rows use combined inputs and macro recall/F1. Literature rows use a different binary KTAS task and the published metric definitions, so this is context, not a like-for-like superiority benchmark.

### Fixed classifier settings

Logistic Regression:  $C = 1$ , balanced training weights, 2,000 maximum iterations. Hist Gradient Boosting: 300 iterations, learning rate 0.1, other settings at the uploaded baseline defaults. Random Forest: 300 trees, minimum leaf size 3,  $\text{max\_features} = 0.7$ , balanced training weights. Random seed is 42. No post-test parameter tuning or probability adjustment is used.

### Data and representation controls

Groups connect repeated normalized complaints or clinical concepts. Numeric imputation/scaling and categorical encoding are fitted on development rows. The full SVD PCA solver is fitted once per input view using development embeddings. The same encoder, labels, rows and classifier settings are retained across the four conditions.

The complete result set contains two input views, two text representations and three classifiers. All twelve configurations and their predictions are saved with input checksums for reproducibility.

[1] Seo et al. (2025). Artificial intelligence for severity triage based on conversations in an emergency department in Korea. Scientific Reports 15, 16870. DOI: 10.1038/s41598-025-99874-0, Table 2. The publication uses binary KTAS and ten-fold cross-validation.